

Separability and the limits of multivariate forecast reconciliation

Supplementary material

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Experiment 2

Table S1 compares separate and joint MinT reconciliation with the base forecasts in Experiment 2. Reconciliation improves substantially on the ARIMA base forecasts, and less on the VAR base forecasts, in every scenario; the difference between separate and joint reconciliation is small in comparison.

Table S1: Experiment 2: MSE of separate and joint MinT reconciliation relative to the base forecasts, averaged over 500 replications, 16 series and horizons 1–12.

Scenario	T	ARIMA		VAR	
		Separate	Joint	Separate	Joint
Separable, $\rho = 0.7$	108	0.941	0.938	0.964	0.966
Separable, $\rho = 0.7$	400	0.905	0.886	0.981	0.980
Separable, $\rho = -0.7$	108	0.939	0.936	0.979	0.983
Separable, $\rho = -0.7$	400	0.896	0.884	0.991	0.990
Variable-specific dynamics	108	0.924	0.930	0.973	0.974
Variable-specific dynamics	400	0.889	0.885	0.977	0.974
Node-varying correlation	108	0.936	0.933	0.958	0.963
Node-varying correlation	400	0.893	0.885	0.982	0.980
Variable-specific node covariance	108	0.944	0.942	0.977	0.980
Variable-specific node covariance	400	0.913	0.905	0.982	0.983
Node-varying dynamics	108	0.939	0.938	0.960	0.960
Node-varying dynamics	400	0.888	0.875	0.959	0.954

Application: ETS base forecasts

Table S2 repeats the accuracy comparison of the main text with base forecasts from automatically selected ETS models. For ETS the covariance matrices are estimated from response residuals, because the innovation residuals of multiplicative-error models are relative errors.

Table S2: Application: MSE relative to base forecasts (ETS), geometric mean over the series at each level, averaged over 48 forecast origins (37 with all twelve horizons) and horizons 1–12.

Method	Admissions			Dismissals		
	Total	Regions	States	Total	Regions	States
Base	1.000	1.000	1.000	1.000	1.000	1.000
OLS	0.965	1.261	1.384	0.929	1.257	1.580
WLS (structural)	0.883	1.076	1.151	0.755	0.953	1.203
MinT, separate	0.809	0.948	0.985	0.627	0.782	0.957
MinT, separate (joint estimate)	0.814	0.951	0.988	0.628	0.785	0.959
MinT, joint	0.769	0.923	0.966	0.637	0.785	0.949

Application: results by horizon

Table S3 breaks down the comparison of joint and separate reconciliation (ARIMA base forecasts) by forecast horizon. For admissions the ratio stays close to one at all horizons. For dismissals, joint reconciliation is worse at short horizons and better at long ones, especially at the national level, so the near-neutral averages in the main text combine gains and losses.

Table S3: Application (ARIMA base forecasts): MSE of joint relative to separate MinT reconciliation by forecast horizon, geometric mean over the series at each level.

Horizon	Admissions			Dismissals		
	Total	Regions	States	Total	Regions	States
1–3	1.004	1.012	1.016	1.111	1.058	1.035
4–6	0.977	0.988	1.005	1.115	1.067	1.048
7–9	1.002	1.007	1.020	0.990	0.999	1.019
10–12	1.026	1.025	1.037	0.840	0.903	0.975

Application: results for every series

Table S4 gives the results for every series. Differences between joint and separate reconciliation are larger for individual states than for the aggregates, and go in both directions.

Table S4: Application (ARIMA base forecasts): MSE of separate and joint MinT reconciliation relative to the base forecasts for every series, averaged over 48 origins (37 with all twelve horizons) and horizons 1–12.

Series	Admissions		Dismissals	
	Separate	Joint	Separate	Joint
Total	0.878	0.880	0.844	0.832
Midwest	1.018	1.014	0.808	0.727
Northeast	1.070	1.095	0.762	0.846
North	0.898	0.916	1.121	1.277
Southeast	0.940	0.939	0.772	0.721
South	0.756	0.756	0.989	0.890
AC	1.048	1.078	0.961	1.000
AL	0.951	0.985	1.014	1.005
AM	1.064	1.054	1.007	1.167

Table S4: Application (ARIMA base forecasts): MSE of separate and joint MinT reconciliation relative to the base forecasts for every series, averaged over 48 origins (37 with all twelve horizons) and horizons 1–12.

Series	Admissions		Dismissals	
	Separate	Joint	Separate	Joint
AP	0.917	0.908	0.933	0.944
BA	1.071	1.093	1.078	1.260
CE	1.029	1.075	0.959	1.068
DF	0.881	0.934	1.015	1.006
ES	0.923	0.926	0.954	0.857
GO	1.018	0.988	0.934	0.823
MA	1.053	1.078	0.802	0.865
MG	0.969	0.945	0.919	0.762
MS	1.140	1.080	1.074	1.074
MT	0.987	1.064	0.852	0.834
PA	0.896	0.922	0.878	1.022
PB	0.944	0.968	0.914	1.012
PE	1.056	1.054	1.088	1.083
PI	0.942	1.010	0.840	0.860
PR	1.026	1.019	0.837	0.761
RJ	0.962	0.987	0.903	1.096
RN	1.094	1.140	0.941	1.061
RO	0.935	1.018	0.929	0.856
RR	0.965	0.970	0.981	1.005
RS	0.890	0.909	0.847	0.787
SC	0.852	0.854	0.986	0.899
SE	1.009	1.025	0.952	1.002
SP	0.997	1.009	0.825	0.720
TO	1.005	1.043	1.051	1.132